Brando Miranda

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Stanford, CA, 94305.
Gates Bldg., 353 Jane Stanford Way

Google Scholar; Website; Stanford Profile; MIT Website

Research Interests

AI for formal mathematics — Lean 4 theorem proving, autoformalization, and end-to-end formal verification of software and mathematics; contamination-resistant evaluation of LLM mathematical reasoning; data-centric machine learning for foundation and frontier models; alternative architectures (e.g. energy-based models) toward Artificial General Intelligence.

Education

Ph.D. in Computer Science — *2022-2026 (expected)*
Stanford University — GPA: 4.045/4.0
Advisor: Prof. Sanmi Koyejo, Stanford Trustworthy AI Research Group (STAIR).

Master of Engineering in Electrical Engineering and Computer Science — *2014-2016*
Massachusetts Institute of Technology — GPA: 4.8/5.0
Advisor: Prof. Tomaso Poggio, Center for Brains, Minds and Machines (CBMM).
Thesis Title: *Function Approximation with Deep Neural and Gaussian Networks.*

Bachelor of Science, Computer Science and Engineering — *2010-2014*
Massachusetts Institute of Technology — Minors: Mathematics, Music

Professional Experience

Stanford University - Stanford, CA — *September 2022 - 2026 (expected)*
Ph.D. Student in Computer Science. Advisor: Professor Sanmi Koyejo

MIT CBMM, UIUC & Stanford — *June 2016 - present*
Research Mentor

Amazon Web Services (AWS) - Cupertino, CA — *June 2024 - September 2024*
Applied Scientist Intern

Morph Labs - Remote — *October 2023 - December 2023*
Machine Learning Research Scientist Consultant

Wise Agents - Stanford Spin-out — *2023*
AI Research Consultant

IBM Research - Yorktown Heights, NY — *May 2022 - August 2022*
Graduate Research Intern

University of Illinois Urbana-Champaign - Urbana-Champaign, IL — *September 2018 - May 2022*
Graduate Student. Advisor: Professor Sanmi Koyejo

IBM Research - Yorktown Heights, NY — *May 2021 - August 2021*
Graduate Research Intern

MIT CBMM (Center for Brain Minds & Machines) - Cambridge, MA — *June 2015 - September 2018*
Research Assistant. Advisor: Professor Tomaso Poggio

Rackspace - San Antonio, TX — *June 2014 - August 2014*
Software Engineering Intern

Adobe - San Jose, CA — *June 2013 - August 2013*
Software Engineering Intern — ML/NLP for Spanish Sentiment Analysis

Publications

Google Scholar metrics (as of May 14, 2026): 3,014+ citations — h-index: 17 — i10-index: 22.

Refereed Publications

(1) E. Chen, A. Gulati, B. Miranda, Z. Tang, S. Koyejo, *Rethinking LLM Judges: Chain-of-Thought and Multi-Step Pipelines for Math Grading.*
(**International Conference on Learning Representations (ICLR) Workshop on Logical Reasoning of Large Language Models. 2026**)

(2) Z. Zhou, X. Lu, C. Cao, B. Miranda, T. Liu, B. Han, S. Koyejo, *CoDaPO: Confidence and Difficulty-Adaptive Policy Optimization for LLM Reasoning.*
(**International Conference on Learning Representations (ICLR) Workshop on Lifelong Agents: Learning, Aligning, Evolving. 2026**)

(3) R. Schaeffer, H. Schoelkopf, B. Miranda, G. Mukobi, V. Madan, A. Ibrahim, H. Bradley, S. Biderman, S. Koyejo, *Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?*
(**International Conference on Machine Learning (ICML). 2025 & International Conference on Machine Learning (ICML) Workshop on Trustworthy Multi-modal Foundation Models and AI Agents (TiFA) — Outstanding Paper Award. 2024**)

(4) A. Gulati, B. Miranda, E. Chen, E. Xia, K. Fronsdal, B. de Moraes Dumont, S. Koyejo, *Putnam-AXIOM: A Functional & Static Benchmark for Measuring Higher Level Mathematical Reasoning in LLMs.*
(**International Conference on Machine Learning (ICML). 2025 & Neural Information Processing Systems (NeurIPS) Workshop on Mathematics and AI (MATH-AI). 2024**)

(5) B. Miranda, Z. Zhou, A. Nie, E. Obbad, L. Aniva, K. Fronsdal, W. Kirk, D. Soylu, et al., *VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4.*
(**2nd Workshop on AI for Math at International Conference on Machine Learning (ICML). 2025**; reviewer recommendation: **Accept (Oral, Top 1)**)

(6) Z. Zhou, X. Lu, C. Cao, B. Miranda, T. Liu, B. Han, S. Koyejo, *CoDaPO: Confidence and Difficulty-Adaptive Policy Optimization for Post-Training Language Models.*
(**2nd Workshop on AI for Math at International Conference on Machine Learning (ICML). 2025**)

(7) L. Aniva, C. Sun, B. Miranda, C. Barrett, S. Koyejo, *Pantograph: A machine-to-machine interaction interface for advanced theorem proving, high level reasoning, and data extraction in Lean 4.*
(**International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS). 2025**)

(8) R. Schaeffer, B. Miranda, J. Kazdan, K. Liu, A. M. Ahmed, N. Mireshghallah, et al., *Causally Quantifying the Effect of Test Set Contamination on Generative Benchmarks.*
(**Neural Information Processing Systems (NeurIPS) Workshop on Evaluating the Evolving LLM Lifecycle. 2025**)

(9) R. Schaeffer, J. Kazdan, Y. Denisov-Blanch, B. Miranda, M. Gerstgrasser, et al., *Position: Machine Learning Conferences Should Establish a 'Refutations and Critiques' Track.*
(**Neural Information Processing Systems (NeurIPS), Main Track. 2025**)

(10) S. Barkallah, S. Daruru, B. Miranda, L. Aniva, A. Nie, S. Koyejo, *VeriBench-FTP: A Formal Theorem Proving Benchmark in Lean 4 for Code Verification.*
(**5th Neural Information Processing Systems (NeurIPS) Workshop on Mathematical Reasoning and AI. 2025**)

(11) R. Schaeffer, K. Liu, B. Miranda, A. M. Ahmed, N. Mireshghallah, S. Koyejo, *The Contamination Paradox: Why Test Set Leakage Can Be Both Potent and Negligible.*
(**Neural Information Processing Systems (NeurIPS) Workshop on Evaluating the Evolving LLM Lifecycle. 2025**)

(12) R. Schaeffer, D. Valentine, L. Bailey, J. Chua, et al., B. Miranda, et al., S. Koyejo, E. Perez, *Failures to Find Transferable Image Jailbreaks Between Vision-Language Models.*
(**International Conference on Learning Representations (ICLR). 2024 & Neural Information Processing Systems (NeurIPS) Workshop on Red Teaming Generative AI, Contributed Talk. 2024**)

(13) R. Schaeffer, M. Khona, S. Chandra, M. Ostrow, B. Miranda, S. Koyejo, *Does Maximizing Neural Regression Scores Teach Us About The Brain?*
(**Neural Information Processing Systems (NeurIPS) Workshop on Unifying Representations in Neural Models (UniReps), 2nd Edition. 2024**)

(14) K. Chawla, A. Sahai, M. DePavia, S. Sundar, B. Miranda, *Quantifying the Importance of Data Alignment in Downstream Model Performance.*
(**International Conference on Learning Representations (ICLR) Workshop on Data-Centric Machine Learning Research (DMLR). 2024**)

(15) A. Gulati, D. Ladsaria, S. Mishra, J. Sidhu, B. Miranda, *An Evaluation Benchmark for Autoformalization in Lean4.*
(**International Conference on Learning Representations (ICLR) Tiny Papers Track, Second Edition. 2024**)

(16) K. Chawla, A. Sahai, M. DePavia, B. Miranda, *A Systematic Study of the Role of Data Quality and Alignment for Fine-tuning LLMs for Enhanced Autoformalization.*
(**International Conference on Learning Representations (ICLR) Tiny Papers Track, Second Edition. 2024**)

(17) R. Schaeffer, B. Miranda, S. Koyejo, *Are Emergent Abilities of Language Models a Mirage?*
(**Neural Information Processing Systems (NeurIPS), Main Track. 2023 — Outstanding Paper Award & Oral**)
Featured in: New York Times, Quanta Magazine, Forbes, Stanford HAI; Cited in the 2024 Economic Report of the President (White House).

(18) B. Miranda*, A. Lee*, P. Yu, S. Koyejo, *Beyond Scale: the Diversity Coefficient as a Data Quality Metric Demonstrates LLMs are Pre-trained on Formally Diverse Data.*
(**International Conference on Machine Learning (ICML) Workshop on Data-Centric Machine Learning & International Conference on Machine Learning (ICML) Workshop on Deployable Generative AI. 2023**)
(*equal contribution)

(19) B. Miranda, P. Yu, Y. Wang, S. Koyejo, *The Curse of Zero Task Diversity: the Failure of Transfer Learning to Outperform MAML and their Empirical Equivalence.*
(**Neural Information Processing Systems (NeurIPS) Workshop on Meta-Learning, Contributed Talk. 2022**)

(20) T. Poggio, A. Banburski, Q. Liao, B. Miranda, L. Rosasco, J. Hidary, *Weight and Batch Normalization implement Classical Generalization Bounds.*
(**International Conference on Machine Learning (ICML) Workshop. 2019**)

(21) D. Kita, B. Miranda, H. Lin, J. Michon, D. Favela, J. Hu, *High-resolution on-chip digital Fourier transform spectroscopy.*
(**Conference on Lasers and Electro-Optics (CLEO): Science and Innovations. 2018**)

(22) D. Kita, B. Miranda, D. Favela, D. Bono, J. Michon, H. Lin, T. Gu, J. Hu, *High-performance and scalable on-chip digital Fourier transform spectroscopy.*
(**Nature Communications 9 (1), 4405. 2018**)

(23) T. Poggio, H. Mhaskar, L. Rosasco, B. Miranda, Q. Liao, *Why and when can deep-but not shallow-networks avoid the curse of dimensionality: A review.*
(**International Journal of Automation and Computing (IJAC), pp. 1-17, 2017. Most Cited Paper Certificate.**)

Preprints and Technical Reports

(1) D. Amrollahi, M. Karimi, B. Miranda, L. Aniva, C. Sun, C. Barrett, S. Koyejo, *AI Coding Benchmarks Need Proofs, Not Just Tests.*
(**Preprint 2026**)

(2) E. Obbad, B. Miranda, D. L. W. Hall, R. Schaeffer, S. Koyejo, P. Liang, *Curating High Quality Pretraining Data for Language Models via Compression Ratios.*
(**Preprint 2026**)

(3) R. Schaeffer, J. Kazdan, B. Abbasi, K. Z. Liu, B. Miranda, A. M. Ahmed, F. Berez, et al., *Quantifying the Effect of Test Set Contamination on Generative Evaluations.*
(**Preprint 2026, arXiv:2601.04301**)

(4) R. Schaeffer, N. Levi, B. Miranda, S. Koyejo, *Pretraining Scaling Laws for Generative Evaluations of Language Models.*
(**Preprint 2025, arXiv:2509.24012**)

(5) R. Schaeffer, N. Levi, A. Kirsch, T. Guenais, B. Miranda, E. Obbad, S. Koyejo, *Evaluating the Robustness of Chinchilla Compute-Optimal Scaling.*
(**Preprint 2025, arXiv:2509.23963**)

(6) W. Chan, M. Souliman, J. Nordhagen, B. Miranda, E. Obbad, S. Koyejo, *Lean-ing on Quality: How High-Quality Data Beats Diverse Multilingual Data in Autoformalization.*
(**Preprint 2025, arXiv:2502.15795**)

(7) Z. Zhou, C. Cao, X. Feng, X. Li, Z. Li, X. Lu, J. Yao, W. Huang, T. Cheng, et al., B. Miranda, *AlphaApollo: Orchestrating Foundation Models and Professional Tools into a Self-Evolving System for Deep Agentic Reasoning.*
(**Preprint 2025, arXiv:2510.06261**)

(8) K. Selva, S. Vittayaareekul, B. Miranda, *Exploring the Efficacy of Meta-Learning: Unveiling Superior Data Diversity Utilization of MAML Over Pre-training.*
(**Preprint 2025, arXiv:2501.08506**)

(9) E. Obbad, I. Mlauzi, B. Miranda, R. Schaeffer, K. Obbad, S. Bedi, S. Koyejo, *ZIP-FIT: Embedding-Free Data Selection via Compression-Based Alignment.*
(**Preprint, arXiv 2024**)

(10) J. Rotella, Z. Qin, A. Z. H. Yang, B. Miranda, M. E. A. Seddik, J. Zuo, H. Hacid, et al., *Synthetic Theorem Generation in Lean.*
(**Preprint 2024**)

(11) B. Miranda (ML Research Scientist Consultant) & Morph Labs Team, *Morph Prover v0 7b: The 1st Frontier Model for the Lean 4 Formal Verification Programming Language.*
[Blog] [Hugging Face Model Card]

(12) B. Miranda, P. Yu, S. Goyal, Y. Wang, S. Koyejo, *Is Pre-training Truly Better Than Meta-Learning*?
(**Preprint 2023**)

(13) B. Miranda, A. Shinnar, V. Pestun, B. Trager, *Transformer Models for Type Inference in the Simply Typed Lambda Calculus: A Case Study in Deep Learning for Code.*
(**Preprint 2023**)

(14) B. Miranda, Y. Wang, S. Koyejo, *Does MAML Only Work via Feature Re-use? A Data Set Centric Perspective.*
(**Preprint 2021**)

(15) B. Miranda, *An Empirical Study of Meta-Learning: a step towards rigorously understanding meta-learning algorithms.*
(**Preprint 2020**)

A. Banburski, Q. Liao, B. Miranda, L. Rosasco, B. Liang, J. Hidary, T. Poggio, Theory III: Dynamics and Generalization in Deep Networks.
(**Preprint 2019**)

Q. Liao, B. Miranda, A. Banburski, J. Hidary, T. Poggio, A Surprising Linear Relationship Predicts Test Performance in Deep Networks.
(**Preprint 2018**)

C. Zhang, Q. Liao, A. Rakhlin, B. Miranda, N. Golowich, T. Poggio, Theory of Deep Learning IIb: Optimization Properties of SGD.
(**Preprint 2018**)

T. Poggio, Q. Liao, B. Miranda, A. Banburski, X. Boix, J. Hidary, Theory IIIb: Generalization in Deep Networks.
(**Preprint 2018**)

T. Poggio, K. Kawaguchi, Q. Liao, B. Miranda, L. Rosasco, X. Boix, J. Hidary, M. Mhaskar, Theory of Deep Learning III: explaining the non-overfitting puzzle.
(**Preprint 2017**)

Awards & Honors

- **ICML 2026 Silver Reviewer** (May 2026) — top reviewer recognition signed by the ICML 2026 Program Chairs.
- **Oral Recommendation (Top 1)**, 2nd AI for Math Workshop @ ICML 2025 — for *VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4* (Miranda et al.); reviewer recommendation "Accept (Oral, Top 1)".
- **Pear AI Researchers Circle** (July 2025) — selective AI-researcher network convened by Pear VC; invited following the final round (R3) of the Pear AI Researcher Grant program.
- **ICML Workshop on Trustworthy Multi-modal Foundation Models and AI Agents (TiFA) — Outstanding Paper Award** (July 2024) — for "Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?" (Schaeffer, Schoelkopf, Miranda, et al.).
- **NeurIPS Outstanding Main Track Paper Award** (December 2023) — top 0.4% of NeurIPS submissions; only 2 main-track papers selected. For "Are Emergent Abilities of Large Language Models a Mirage?" (Schaeffer, Miranda, Koyejo).
- **EDGE Scholar**, Stanford University (September 2022) — Stanford fellowship supporting first-generation / low-income PhD students with additional mentorship and stipend.
- **Stanford School of Engineering Fellowship** (September 2022) — multi-year departmental fellowship for incoming Stanford engineering PhD students.
- **Honorable Mention, Ford Foundation Predoctoral Fellowship** (2020, 2021) — national fellowship recognizing PhD applicants with potential to diversify the U.S. professoriate.
- **Best Research Project Award**, UIUC graduate course CS 598 "Learning to Learn" (Prof. Y. Wang, December 2020).
- **HSF Scholar**, Hispanic Scholarship Fund (2020) — competitive national merit fellowship for Hispanic graduate students.
- **Most Cited Paper Certificate**, International Journal of Automation & Computing (IJAC, December 2019) — for "Why and when can deep- but not shallow-networks avoid the curse of dimensionality: a review" (Poggio, Mhaskar, Rosasco, Miranda, Liao).
- **Computer Science Excellence Saburo Muroga Endowed Fellowship**, UIUC (2019-2020) — top departmental fellowship for outstanding incoming CS PhD students.
- **Sloan Scholar**, Alfred P. Sloan Foundation Minority Ph.D. (MPHD) Program (2018-2019) — prestigious national fellowship supporting underrepresented STEM PhD students.
- **Grainger Engineering SURGE Fellowship**, UIUC (2018-2019) — multi-year college-level fellowship for diverse engineering PhD students.
- **Chopper Trading, LLC, Best Strategy Report Award**, MIT BattleCode AI competition (2013) — top finisher among ∼300 teams.
- **MIT Mitchell B. Kaufman Memorial Scholarship** (2012-2013, 2013-2014).
- **MIT Eugene and Margaret McDermott Scholarship** (2012-2013, 2013-2014).
- **Greengates Scholarship** (30% 2007, 50% 2008, 100% 2009, 100% 2010).
- **High Achievement Prize Award**, Greengates School (2007, 2008, 2009, 2010) — equivalent to Valedictorian.
- **Best All-Round Student Award**, Greengates School (2010).
- **Achievement Prize Award**, Greengates School (2006).
- **Certificate for Progress Award**, Greengates School (2005).

Media Coverage

- **Hacker News / Y Combinator (2025)**: Putnam-AXIOM benchmark posted on Hacker News — our *Putnam-AXIOM* benchmark for higher-level LLM mathematical reasoning drew front-page attention on Y Combinator's influential tech-news forum
- **White House Economic Report of the President (2024)**: Our work on emergent abilities was cited in the 2024 Economic Report of the President – the White House's annual flagship document on national economic policy
- **Quanta Magazine (2024)**: "How Quickly Do Large Language Models Learn Unexpected Skills?"
- **Andrew Ng (2024)**: Endorsed our paper as evidence that AGI development will be smooth and predictable
- **The New York Times (2023)**: "Silicon Valley Confronts the Idea That the 'Singularity' Is Here"
- **Forbes (2023)**: "AI 'Emergent Abilities' Are A Mirage, Says AI Researcher"
- **AK / @_akhaliq on X (2023)**: AK shared our *Beyond Scale* paper on X — the influential AI-research-paper aggregator (Gradio founder; now at HuggingFace) tweeted our *Beyond Scale* data-diversity paper to his large research audience (68.9K views)

- **Stanford AI Lab Blog (ICML 2023)**: Our *Beyond Scale* and *Is Pre-training Truly Better Than Meta-Learning?* papers were featured in the Stanford AI Lab blog roundup of ICML 2023
- **Additional coverage**: Stanford HAI, Y Combinator News, Vice, Medium, HackerNews, NeurIPS blog, Reddit

Invited & Contributed Talks

Applications of Trustworthy Machine Reasoning with AI Coding Agents
AAAI 2026 Tutorial "Trustworthy Machine Reasoning with Foundation Models" (Part IV, 50 min). Co-presented with B. Han, Z. Zhou, C. Cao, P. Lu, S. Koyejo. *Singapore, January 20, 2026.*

Emergent Abilities in Large Language Models: Mirage or an Elusive Predictive Frontier?
Hong Kong Baptist University, Department of Computer Science Seminar (2026 Series). *Hong Kong, January 19, 2026.*

VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4
Lawrence Livermore National Laboratory Reading Group (invited). *July 31, 2025.*

Intro to Lean 4, VeriBench, and Trustworthy Testing of AI-Generated Code
Prof. Azalia Mirhoseini's Lab Meeting, Stanford (invited).

Failures to Find Transferable Image Jailbreaks Between Vision-Language Models
NeurIPS Red Teaming GenAI Workshop, Contributed Talk. *December 2024.*

Are Emergent Abilities of Large Language Models a Mirage?
Stanford IEEE Invited Talk. *Stanford, CA, 2023.*

Are Emergent Abilities of Large Language Models a Mirage? — Why Has Predicting Downstream Capabilities Remained Elusive?
Lawrence Livermore National Laboratory (invited talk). *2023.*

Emergent Abilities of Large Language Models
Amazon Research (invited talk). *2023.*

The Curse of Low Task Diversity: On the Failure of Transfer Learning to Outperform MAML and Their Empirical Equivalence
NeurIPS Meta-Learning Workshop, Contributed Talk. *New Orleans, LA, December 2022.*

Selected Writing

- **Asking the Right Question: Formal Methods as Scalable Oversight** *April 2026*
- **What I Learned Building a Correctness-Gated Multi-Agent Workflow for Research** *April 2026*

Academic Service

- **NeurIPS 2026** (Neural Information Processing Systems) Conference reviewer
- **ICLR 2026** (International Conference on Learning Representations) Conference reviewer
- **ICML 2026** (International Conference on Machine Learning) Conference reviewer – **Silver Reviewer designation** (top reviewer recognition)
- **ICLR 2025** (International Conference on Learning Representations) Conference reviewer
- **ICML 2025** 2nd AI for Math (AI4MATH) Workshop reviewer
- **ICLR 2024** Data-centric Machine Learning Research (DMLR) Workshop reviewer
- **NeurIPS 2023** Mathematical Reasoning and AI (MATH-AI) Workshop reviewer
- **ICLR 2020** (International Conference on Learning Representations) Conference reviewer
- **JMLR 2018** (Journal of Machine Learning Research) joint review with Qianli Liao

Service & Outreach

- Graduate advisor for Latinos in Computer Science (LCS) UIUC 2019-Present
- Outreach research mentorship Distributed Research Experiences for Undergraduates (DREU) UIUC 2019
- Outreach research mentorship Undergraduate Research Opportunity Program (UROP) MIT 2017-2018
- Outreach research mentorship Engineering of Intelligence Team (EIT) CBMM MIT 2017-2018

Stanford University - Stanford, CA *2023 - 2025*
Course Assistant (3 quarters) & Instructor of Record (Spring)

- CS 197 *How to Do CS Research* with Prof. Michael Bernstein — TA'd 3 quarters; served as **Instructor of Record** for the Spring offering
- Mentored undergraduate research projects from the course that produced **2 workshop publications** (including ICML workshop tracks)

UIUC - Urbana-Champaign, IL *August 2020 - December 2020*
Graduate Teaching Assistant

- CS 446 Machine Learning
- Responsible for problem sets and exam design. Held weekly office hours.

MIT CBMM - Cambridge, MA *September 2016 - December 2016*
Graduate Teaching Assistant

- Statistical Learning Theory & Applications (9.520/6.860) with professor Lorenzo Rosasco & Tomaso Poggio
- Made the first draft of course slides for MIT's OpenCourseWare (OCW)
- Graded final projects
- Provided extra support for office hours

MIT EECS - Cambridge, MA *February 2016 - June 2016*
Graduate Teaching Assistant

- Introduction to Algorithms (6.006) with professor Nancy Lynch, Bruce Tidor and Aleksander Madry
- Held weekly office hours
- Helped prepare problem sets and exams
- Graded exams

MIT EECS - Cambridge, MA *September 2015 - December 2015*
Graduate Teaching Assistant

- Design & Analysis of Algorithms (6.046) with Shafi Goldwasser, Dana Moshkovitz, Nir N. Shavit
- Held weekly office hours
- Instructed students across weekly recitations
- Helped prepare problem sets and exams

MIT EECS - Cambridge, MA *February 2015 - May 2015*
Graduate Teaching Assistant

- Introduction to Machine Learning (6.036) with professor Tommi S Jaakkola, Suvrit Sra & Regina Barzilay
- Instructed students across bi-weekly recitations
- Held weekly office hours
- Helped prepare problem sets, projects and exams
- Graded problem sets, projects and exams

MIT EECS - Cambridge, MA *September 2014 - December 2014*
Graduate Teaching Assistant

- Mathematics for Computer Science (6.042) with professor F.Thomson Leighton & Ankur Moitra
- Instructed students across four recitations and problem solving sessions weekly
- Held weekly office hours
- Graded exams

MIT - Cambridge, MA *September 2012 - June 2013*
National Honor Society for Computer Science and Electrical Engineering (Eta Kappa Nu)

- Algorithms tutor for undergraduates at MIT

Certification

- Coursera: Learning how to Learn: Powerful mental tools to help you master tough subjects (with Honors), awarded by UC San Diego, Dr. Barbara Oakley, Dr. Terrence Sejnowski & Linda Walker
- Coursera: Mindshift: Break Through Obstacles to Learning and Discover Your Hidden Potential (with Honors), by McMaster University Dr. Barbara Oakley, Dr. Terrence Sejnowski & Linda Walker
- Certified Dance instructor with the World-Mastery program: completed the online training by world renowned artists Korke & Judith

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Lean 4, OCaml, Coq, Java, Go, Javascript

ML/AI Tools: Hugging Face, Weights & Biases, DSPy, git, UNIX, vim

Languages: English, Spanish (both native fluency)

Leadership

Stanford AI for Lean (Lean AI Club) - Stanford, CA *2025 - Present*
Founding Member — Stanford Lean AI Club (also aiforlean.org), a research community advancing AI for Lean theorem proving and formalizing mathematics.

Stanford Bachata Sensual & Brazilian Zouk (SBSBZ) - Stanford, CA *September 2021 - Present*
Founder & President — founded the first Bachata Sensual & Zouk student organization at Stanford; 200+ members; open-access lesson summaries on YouTube.

UIUC Latinos in Computer Science (LCS) - Urbana-Champaign, IL *December 2019 - Present*
Graduate Advisor — founded the LCS Professional & Wellness Development Colloquium speaker series.

UIUC Bachata Sensual & Zouk (Latin Dance Group) - Urbana-Champaign, IL *January 2019 - December 2021*
Founder & President — founded the first Bachata Sensual & Zouk group at UIUC. **Bachata Sensual Excellence Diploma**, Korke & Judith *World Mastery* (Beginner, Intermediate, Advanced), 2019.

Activities

MIT Chamber Music Society - Cambridge, MA *September 2011 - June 2014*
Jazz Musician

- Alto Saxophone player with MIT's Jazz combo as part of the Chamber Music Society lead by Boston jazz artist Keala Kaumeheiwa
- Performed, Composed and arranged several Jazz pieces

MIT Wind Ensemble (MITWE) - Cambridge, MA *February 2011 - June 2011*
Clarinetist

- Clarinet player with director Dr. Harris
- Performed with the Wind Ensemble at the MIT 150th Convocation

Family Background

Latino/Hispanic, Mexican heritage.